

Computer Networks Spring 2026

Assignment#2 (6A & 6B) Solution

Question: 01

- a) The total amount of time to get the IP address from the DNSs =
 $20+10+25+15+30 = 100 \text{ ms}$
After getting the IP address, two RTTs are needed to receive the file since one RTT is used to set up the TCP connection and another RTT is used to request and receive the file.
Delay in receiving the file = 2 RTTs = $2 \times 20 = 40\text{ms}$
Therefore, the total response time = $100 \text{ ms} + 40 \text{ ms} = 140 \text{ ms}$
- b) (i) Total response time = time to get IP address from server + time in receiving base HTML file + time in receiving 5 small objects = $100 \text{ ms} + 40 \text{ ms} + 5 \times (2 \times 20) = 340 \text{ ms}$
(ii) Total response time = time to get IP address from server + time in receiving base HTML file + time in receiving 5 small objects = $100 \text{ ms} + 40 \text{ ms} + 2 \times (2 \times 20) = 220 \text{ ms}$
Note: Request 3 files concurrently and then request 2 files concurrently.
(iii) Total response time = time to get IP address from server + time in receiving base HTML file + time in receiving 5 small objects without additional request = $100 \text{ ms} + 40 \text{ ms} + 20 = 160 \text{ ms}$

Question 2:

- a) No records are to be added to the root name servers.
- b) The following RRs need to be added to a TLD server:
(fastburgers.com, authoritative.fastburgers.com, NS),
(fastburger.com, authoritative.fastburgers.com, NS),
(authoritative.fastburgers.com, 20.21.22.25, A)
These records will usually be added to all .com TLD name servers by the DNS Registrar company.
- c) The following RRs need to be added to your authoritative name server:
(fastburger.com, fastburgers.com, CNAME),
(fastburgers.com, 20.21.22.23, A),
(fastburgers.com, smtp.fastburgers.com, MX),
(smtp.fastburgers.com, 20.21.22.24, A),
database.fastburgers.com, 20.21.22.26, A)

Question 3:

- i. No, since the response code is 404 Not Found
- ii. Non persistent (Refer to Connection: Close)
- iii. text/html
- iv. 258 bytes
- v. Apache/2.2.3

Question 4:

a. Utilization:

- Traffic In = $100 \text{ req/s} \times 100 \text{ kbits/req} = 10,000 \text{ kbps} = 10 \text{ Mbps}$.
- Link Cap = 15 Mbps.
- Utilization = $10 / 15 = 0.66$ (or 66.7%).

b. With Cache (0.4 hit rate):

- 40% satisfied locally. 60% go to origin.
- New Arrival Rate to link = $0.6 \times 100 = 60 \text{ req/s}$.
- Traffic = $60 \times 100 \text{ kb} = 6 \text{ Mbps}$.
- New Utilization = $6 / 15 = 0.4$ (or 40%).

c. End-to-End Delay:

- Delay_total = $(0.4 \times 0.01\text{s}) + (0.6 \times 2.0\text{s})$.
- Delay = $0.004 + 1.2 = 1.204 \text{ seconds}$.
- *(Note: 2.0s includes the internet RTT; since access link utilization is low (0.4), queuing delay is ignored as per instructions).*